

Science Olympiad Codebusters C Purdue Invitational

January 18th, 2025



Instructions:

- You will have **50** minutes to complete this test
- You may bring up to **(3)** 4 or 5 function calculators, if you have any additional please place it underneath your seat
- Please make sure that all answers are **clearly visible** for the grader to score and boxed when requested.
- You may take apart the test as you wish, however you should write your team number **on top of every page** and a **250** point deduction will be taken if the pages are not put back in the correct order
- The first question on the exam is **Timed**. If you believe you have solved the question in the first 10 minutes, signal to the event supervisor who will come over to check it. If you are correct, your time will be recorded and a bonus issued; if you have more than 2 errors you will be notified and allowed to retry for a bonus throughout the first 10 minutes.
- If you successfully complete the timed question after 10 minutes, you may still get the base points for the question however it will not be checked and no bonus will be awarded.
- As stated in Section 4.f, there are 3 questions marked for special bonus: **#11, #16, #21**
- Page 2 contains scoring information. Page 3 (the front of the 2nd sheet) is the timed question.

School/Team Name: _____

Team Number: _____

Written by:

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Thank you to Aneesh Iyer (Mason '25) for the template for this test!

Scoring Tables

For Event Supervisor use only; only write team number on this page

Question	Points	Errors	Total		Question	Points	Errors	Total
Timed	250				12	200		
1	265				13	225		
2	150				14	100		
3	150				15	575		
4	255				16★	400		
5	255				17	150		
6	300				18	555		
7	200				19	200		
8	350				20	600		
9	645				21★	250		
10	250				22	300		
11★	325							
Total	22 Questions + Timed				XXXX Points			

Test Score (sum of both “Total” columns above): _____

Timed Question:

Time to solve question: _____

M:SS

Timed Bonus: $(600 - \text{Time in Seconds}) * 2 = \text{Timed Bonus}$

Time in Seconds

Timed Bonus

Special Bonus:

Awarded for full points on marked questions (circle one)

- 0 correct: 0 points
- 1 correct: 150 points
- 2 correct: 400 points
- 3 correct: 750 points

Final Score: _____ + _____ + _____ = _____
 Test Score Timed Bonus Special Bonus Final Score

Placement: _____

Reference Materials:

Keys	A	B	C	D	E	F	G	H	I	J	K	L	M
A,B	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
C,D	O	P	Q	R	S	T	U	V	W	X	Y	Z	N
E,F	P	Q	R	S	T	U	V	W	X	Y	Z	N	O
G,H	Q	R	S	T	U	V	W	X	Y	Z	N	O	P
I,J	R	S	T	U	V	W	X	Y	Z	N	O	P	Q
K,L	S	T	U	V	W	X	Y	Z	N	O	P	Q	R
M,N	T	U	V	W	X	Y	Z	N	O	P	Q	R	S
O,P	U	V	W	X	Y	Z	N	O	P	Q	R	S	T
Q,R	V	W	X	Y	Z	N	O	P	Q	R	S	T	U
S,T	W	X	Y	Z	N	O	P	Q	R	S	T	U	V
U,V	X	Y	Z	N	O	P	Q	R	S	T	U	V	W
W,X	Y	Z	N	O	P	Q	R	S	T	U	V	W	X
Y,Z	Z	N	O	P	Q	R	S	T	U	V	W	X	Y

AAAAA	A
AAAAB	B
AAABA	C
AAABB	D
AABAA	E
AABAB	F
AABBA	G
AABBB	H
ABAAA	I/J
ABAAB	K
ABABA	L
ABABB	M

ABBAA	N
ABBAB	O
ABBBA	P
ABBBB	Q
BAAAA	R
BAAAB	S
BAABA	T
BAABB	U/V
BABAA	W
BABAB	X
BABBA	Y
BABBB	Z

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

Frequency of English Letters:

E - 12.51%	S - 6.54%	C - 3.06%	G - 1.96%	K - 0.67%
T - 9.25%	R - 6.12%	U - 2.71%	W - 1.92%	X - 0.19%
A - 8.04%	H - 5.49%	M - 2.53%	Y - 1.73%	J - 0.16%
O - 7.60%	L - 4.14%	F - 2.30%	B - 1.54%	Q - 0.11%
I - 7.26%	D - 3.99%	P - 2.00%	V - 0.99%	Z - 0.09%
N - 7.09%				

Frequency of Spanish Letters:

E - 14.08%	I - 5.98%	M - 3.08%	Y - 1.09%	Z - 0.47%
A - 12.16%	L - 5.24%	P - 2.89%	V - 1.05%	N - 0.17%
O - 9.20%	U - 4.69%	B - 1.49%	G - 1.00%	X - 0.14%
S - 7.20%	D - 4.67%	H - 1.18%	F - 0.69%	K - 0.11%
N - 6.83%	T - 4.60%	Q - 1.11%	J - 0.52%	W - 0.04%
R - 6.41%	C - 3.87%			

For the purposes of cryptograms it is customary to treat n and ñ as distinct letters, but a and á are the same letter. Likewise for e and é, and i and í. In other words, all the accent marks get amputated when working with cryptograms. Also, while some older Spanish dictionaries consider ch, ll, and rr, to be their own letters—this has fallen out of modern usage. Accordingly, “burro” is considered as five letters: “b-u-r-r-o” and not as four letters “b-u-rr-o.”

Morse Code:

A	•—
B	—•••
C	—•—•
D	—••
E	•

F	••—•
G	—••
H	••••
I	••
J	•----

K	—•—
L	•—••
M	—
N	—•
O	---

P	•--•
Q	--•—
R	•—•
S	••••
T	—

U	••—
V	•••—
W	•---
X	--••
Y	—•---
Z	--••

0	-----
1	•-----

2	••----
3	•••---

4	••••—
5	•••••

6	—••••
7	--•••

8	----••
9	-----•

•	E
••	I
•••	S
••••	H
—•••	B
-----	0
•••••	5

—	T
•—	A
••—	U
•••—	V
—••—	X
•-----	1
—••••	6

—•	N
•—•	R
••—•	F
—•—•	C
••---	2
---•••	7

--	M
•--	W
•••	L
—••	Y
•••---	3
---••	8

—••	D
•---•	P
---••	Z
••••—	4
-----•	9

—•—	K
•---	J
—•—	Q

---•	G
---	O

[250] **Timed Question:** Decode the following **Aristocrat** which is a quotation from Purdue alum Neil Armstrong. When you complete it, yell out “**I JUST VENTED**” so that your answer can be verified and time recorded by the supervisor.

LZ FCXXMJGB FZKWH OM ZSEZ ZSEZ ZLJB AME, AKMZZB EJX

VGCM, QEF ZSM MEKZS. L XLXJ'Z UMMG GLHM E NLEJZ. L

UMGZ RMKB, RMKB FOEGG.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency:	2	5	3		9	4	7	2		5	5	7	13	1	2		1	2	4		2	1	1	5		14
Replacement:																										

[265] **Question 1:** Solve this **Aristocrat** which is a quote from the movie *Now You See Me*.

NK AKBC ADKNC, SCO FDD KGCY BC MCAFRNC OQC ADKNCY

PKR OQXTU PKR FYC, OQC DCNN PKR'DD FAORFDDP NCC.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency:	5	2	14	9		5	1				8		1	7	5	4	3	5	1	1	1			1	3	
Replacement:																										

[150] **Question 2:** Solve this **Baconian** which is a quote from *Avatar: The Last Airbender*.

ABBBAAAAAABBAABAABABAAABAAAAABAAAAABAAAAAABBAABA

AAABABABAABBBAAABABAAAABBBABBBAAAAAABBAAAAABBBAAABA

BBABABAAABAAABAAABBAABAAAAAABAABAAABBB

[150] **Question 3:** Solve this **Nihilist** cipher with keyword **ROBERT** and polybius key **FROST**. It is a quote from a famous poet.

45 55 37 57 24 40 37 65 35 37 36 29 45

36 43 67 26 60 53 58 65

50 63 40 24 67 37 57 45 57 43 28 54 46 27

47 33 56 65 50 54 29 45 55 57 58 23 40 ; 45 28

53 38 37 29 25 55

[255] **Question 4:** Solve this **Aristocrat** which is an iconic quote from *Game of Thrones*.

FQSRH LHV'U S OLU. FQSRH LH S WSJJCP. ASVK EQR UPK

UR FWLAD LU BSLW, SVJ VCZCP ICU UR UPK SISLV. UQC

BSWW DPCSXH UQCA. SVJ HRAC SPC ILZCV S FQSVFC UR

FWLAD, DYU UQCK PCBYHC.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency:	5	3	14	4	1	6		7	3	4	4	9			1	7	7	7	16		13	8	6	1	2	2
Replacement:																										

[255] **Question 5:** Find the keyword that was used to create this **Aristocrat** which is encoded using a **K1** alphabet. You are told that the keyword is **6 letters** long. **Box** your keyword.

RP TBJX BH RP QBHD BR STT, PRAXQKBHX MPI'QX CIHR SF

BFXQR VAIFD PY QSFWPETM SHHXEUTXW EPTXVITXH WQBYRBFZ

KAXQXJXQ RAX IFBJXQHX UTPKH MPI.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency:	4	9	1	2	3	6		9	6	3	3		3			9	9	8	4	8	2	2	3	15	2	1
Replacement:																										

[300] **Question 6:** Solve this **Baconian** which is a quote from my favorite movie, *Megamind*.

BAECIOUDFGYAHEIJOUYAEIKOUYLAEIMOUYNAPQEIORSUTYAVEIOUY

AEWIOUXZYABEIOCUYDAEIOUYAEFGIOHJUKLYAEMNIOPUQRYSAETIO

UVWXYAZEIBOUYAEICOUYDAEIFOUYGAHJEIOKLUMNYAPEIQROSUTVY

WXAEZIOUBCDYAFEGIOUYAEHIOUJYAEIKOLMUYAEIOUNPYAEIOQRSU

YTAEIVWXOUZYABEIOUYACEIODUYAFEIOGUHJYAEKLIMONPQRSUYTV

AEWIOUYXAEIZBOUYCDAFEIGOYUJAEIKLOUYAMNEIPOQUYAEIORS

YAEIOTV

[200] **Question 7:** Solve this **Hill** cipher which is a quote from Anakin Skywalker from *Star Wars*. The keyword is HUNT.

$$\begin{pmatrix} H & U \\ N & T \end{pmatrix} \equiv \begin{pmatrix} 7 & 20 \\ 13 & 19 \end{pmatrix}$$

N Q A E A E I D E L S X Z L T D H G S K E N C H

[350] **Question 8:** Solve this **Nihilist** cipher with keyword **DERMATOGLYPHICS** and polybius key **UNCOPYRIGHTABLE**.

72 60 57 59 46 65 35 57 57 32 66 57 35 54

82 66 70 65 79 53 84 28 46 75 53 37 60

54 38 86 75 49 34 68 67 82 45 77 48 43 56 76 46 25

82 66 70 57 56 56 65 37 75 59 55 47 37 47 24 83 65 70

75 67 63 56 28 35 65 43 50 40 58 45 82 64 47 46 76

83 54 26 48 68 56 49 60 54 44 86 63 .

[645] **Question 9:** Solve this **Patristocrat** that is a quotation from *Ted Lasso*.

CASLP CBMZL VEGMS MCBBL PMEGC JDNLS BMLVE EGMSM

EGRJK MCJDB RYRJY EGREB KMCJD RTLJM RJYKM CJDBR

YRCJE JLLJM CJEGC BSLLP RTLJM

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency:	1	8	10	4	9		6			13	3	11	13	1		3		9	5	2		2			4	1
Replacement:																										

[250] **Question 10:** Solve this **Hill** cipher using the keyword **BALL**. It is a quote from Joe Namath.

$$\begin{pmatrix} B & A \\ L & L \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 11 & 11 \end{pmatrix}$$

I N Y C U M R X N O G M I X G O L I T A E A A E W H Y S O Y T B L I

[325] **Question 11:** Solve this **Aristocrat** that has some marginal spelling errors. ★ **Special Bonus Question** ★

VLTA UDYC YC VLTU UDK KLB. YU YC LTU KMKL UDK

GKNYLYLN TO UDK KLB. GWUU YU YC, QKJDPQC, UDK

KLB TO UDK GKNYLLYLN.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency:	1	3	5	7			3			1	13	12	1	4	2	1	2			5	12	2	1		10	
Replacement:																										

[200] **Question 12:** Solve this Albert Einstein quote encrypted using a **Porta** cipher and a keyword of **SARCASM**.

ROZZV NCNGB AGQOE TKOEN OESQM VKXKU ZGARC NEKSN

EWQHU OAJAL CQRVK FWAYW ZEIKP GNCUX KRTKF LLNNW

AHKAQ RMMVM XIFZ

[225] **Question 13:** Solve this Lebron James quote encrypted using a **Porta** cipher and a keyword of **GOAT**.

Y XBED RAFG OBA DUYU Y UZ FB OBA PAPQ Y

CRRWO, ONSUHJU P QFK'M JWKM NEIVBZI MB

TKHJ VI PQNKMVKI. P'Z UYRR W CACNBORIL.

EBL SUA YQSY VU VNJNYGXQSY VQG.

[100] **Question 14:** Solve this **Cryptarithm** using the equation below. The solution is microwave-related.

Values to decode for solution

2 8 1 9 9 2 8 0 3 1 9 8

--	--	--	--	--	--	--	--	--	--	--	--

Cryptarithm formula

$$\begin{array}{r}
 \text{E A T} \\
 + \text{T H A T} \\
 \hline
 \text{A P P L E}
 \end{array}$$

	0	1	2	3	4	5	6	7	8	9
A										
E										
H										
L										
P										
T										

SQHMC WSEMA BFLNU LENWF GSJML E

[illegible][illegible]

[150] **Question 17:** Solve this **Complete Columnar** cipher with crib **LUDICROUS**. The quote is from *SPACEBALLS*, the greatest science fiction parody of all time.

PSECD IIWOS ELTRX TOADE EOOX GSELS SOVIE LDWGU

HTHUP

[555] **Question 18:** Solve this **K1 Aristocrat** that has major spelling errors and a keyword of **PATHS**.

IXX ZXHUNK NRBOZQON RW H CXXN, HWN R— R IXXT IJO CXW

UOKK IZOBQUON LEO. HWN IJHI JHK VHRN HUU IJO

NRPPZOWMO.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency:		2	2		1			8	7	4	4	1	1	9	11	2	1	6		1	5	1	5	8		4
Replacement:																										

[200] **Question 19:** Solve this **Complete Columnar** cipher with an odd number of columns and the crib **CHIMNEY**. The solution is a quote from *Mary Poppins*.

RHLRD STBEN IEEOD FWTBI SDMES LAEHO URTTN PHEOO

AAEHT SCSHW YTGET SREYX TETEO STDAH ESEWR UNTIE

HAHW

[600] **Question 20:** Solve this **Spanish Aristocrat** that is from an insane movie that I watched in class in 9th grade. The last letter decodes to an **R**.

UR IOUR SK UURÑOÑ ZBODSR BDR LVZO ZKHUUO DR KF KU

FVILUK NKZNR SK UURÑOÑ. FVDR MBK O JKZKF BDR KILVKYO

GO DR LBKSK LOÑOÑ

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		5		6		4	1	1	3	1	14	5	1	2	6	11			12	4		10	4			1	5
Replacement																											

[250] **Question 21:** Solve this **Fractionated Morse** cipher with keyword **WISTFUL**, given that the first three letters of the plaintext are **I AM**. The solution is a quote from Oscar Wilde. ★ **Special Bonus Question** ★

S P V Y S H Z T P S L I Q T

Z O S U N W V M K X S J L S O

Z S H R Z Y U E C Q T O A P R R B

U O L A A E T L Y J H P A S V K T

Y J W L M Y B U J O L M F O R V B

Replacement:																											
	•	•	•	•	•	•	•	•	•	—	—	—	—	—	—	—	—	×	×	×	×	×	×	×	×	×	×
	•	•	•	—	—	—	×	×	×	•	•	•	—	—	—	×	×	×	•	•	•	—	—	—	×	×	×
	•	—	×	•	—	×	•	—	×	•	—	×	•	—	×	•	—	×	•	—	×	•	—	×	•	—	—

E R K F I V M X G S R D S I

[illegible]